

What does a language-model supervisor contribute to a learning controller?

A controlled study of agentic reward and hyper-parameter supervision for residual pitch control of a wind turbine

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Abstract

Language-model agents now write and tune the rewards of reinforcement-learning controllers. We ask what such an agent contributes when the controller is a bounded residual on an engineered pitch controller of the NREL 5 MW turbine, and we answer with one protocol run to the end: every 30 training episodes the agent proposes candidates, each is verified by a 30-episode fork, the best continues, and a guardrail rolls back; three levers (reward weights, PPO hyper-parameters, the reward expression itself) each get a random-search control; the loop is ablated (agent asked once and applied blind; agent told an objective other than the one that selects); the whole study is repeated on two base controllers, the gain-scheduled PI and a linear-parameter-varying MPC that is the strongest controller on the objective, exact and with a 5 % error in its aerodynamic model; every number is a seed-paired held-out evaluation on two disjoint wind sets with exact permutation tests. On a tuned default the numeric levers are worth nothing, for the agent and for random search alike. The reward expression is the one lever that changes what a solution is made of: the agent’s rewards reach a balanced solution, every metric improved including tower fatigue, in 3 of 5 seeds where no fixed-reward or weight-searched run reaches it in any, but they do not change the level of the objective (+0.4 / +0.5 points, within seed noise). A random draw from the agent’s own vocabulary of term shapes, verified by the same loop, reaches the same composition; the agent without the loop is the fixed reward; the agent told the wrong objective, selected by the right one, is as good as the agent told the right one. On the MPC base the residual has little left to add (+0.3 with the fixed reward, +1.0 with the agent’s) and, when the MPC’s model is 5 % wrong, repairs it; the agent’s reward is again not distinguishable from the fixed one. The agent’s contribution is a vocabulary of reward structures, made real by verification against the true objective—not a search, not a prompt, and not a level. Four protocol defects, each worth more than any agent, are documented as rules for such studies.

1 Introduction

Reinforcement learning (RL) has been proposed for wind-turbine pitch control as a model-free alternative to model predictive control (MPC) [1–4], and its reward is its design problem. Language-model (LLM) agents are the newest answer to that problem: Eureka evolves reward code against a task score [5], Language-to-Rewards and Text2Reward map instructions to reward parameters and programs [6, 7], DrEureka adds domain-randomisation design [8], and agentic reward designers close the loop with semantic feedback [9]. These systems are evaluated by the policy they produce. What the agent itself contributes—beyond a search that any verification loop could run, and beyond the prior a control engineer would write down—is rarely separated out, and for a physical plant with a strong engineered baseline the answer decides whether the agent belongs in the loop at all.

We separate it out on one benchmark. The controller is a bounded residual on top of a base pitch controller of the NREL 5 MW turbine in OpenFAST, trained by PPO; the objective is the

metric set of a published RL pitch-control study (power MSE, generator-speed MSE, tower-base and blade-root fatigue, paired against the gain-scheduled PI); the agent supervises training every 30 episodes. The study is designed so that each of the agent’s possible contributions has a control: three *levers* (reward weights, hyper-parameters, reward expression), each with a random-search control of the same shape; the verification *loop* ablated (asked once, no fork; told a different objective than the one that selects); the agent’s *prior* ablated (a random structural search over expressions drawn from the agent’s own vocabulary of term shapes, passed through the same loop); and two *base controllers*, the PI and the strongest controller on the objective, an LPV-MPC with a correctly scaled cost, exact and with a 5 % model error, so that the agent is tested where the residual has much to do and where it has little.

Findings. Numeric levers are worth nothing on a tuned default, whoever proposes. The reward expression changes the composition of the solution—the agent’s rewards reach a balanced solution with every metric improved, including tower fatigue, in 3 of 5 seeds, where no fixed-reward or weight-searched run reaches it in any—but not the level of the objective. The loop is necessary and the prompt is not: asked once, the agent produces the fixed reward; told the wrong objective and selected by the right one, it produces the balanced solution as often. A random draw from the agent’s vocabulary, verified by the same loop, reaches the same composition. On the MPC base the residual adds +0.3 (fixed reward) and +1.0 (agent) to a controller that is already positive on every metric; when the MPC’s aerodynamic model is 5 % wrong the residual repairs it and the agent’s reward is once more indistinguishable from the fixed one. Along the way four protocol defects were found, each moving the comparison by more than any agent: the subset on which the reward’s speed term is gated, the scale of the critic’s target, the scale of the MPC’s cost terms, and the channel through which a reference controller is evaluated.

Contributions. (a) A controlled protocol for agentic supervision of a learning controller—levers with matched random controls, loop and prior ablations, two bases, seed pairing, exact permutation tests—that any such study can reuse. (b) Its outcome: the agent’s contribution is a vocabulary of reward structures realised by verification against the true objective, with no measurable effect on the level of the objective on either base. (c) Design and protocol rules distilled from the defects that mattered more than the agent.

2 Setup

2.1 Plant, objective, statistics

NREL 5 MW onshore turbine [10] in OpenFAST 4.2.1 [11], controlled by ROSCO 2.10.5 [12] through a patched ZeroMQ interface at the 10 ms control step; the ROSCO configuration is the gain-scheduled PI (GSPI) with peak shaving on and every other load-mitigation feature off, constant torque above rated, and it is the pairing reference for every metric. Wind: TurbSim, 8, 12.5 and 15 m/s mean hub wind, turbulence intensity 8 %, 150 s episodes of which 130 s are scored. Realisations with TurbSim seeds 1–2 are the supervisor’s evaluation winds; seeds 3–6 and 7–10 are two disjoint held-out sets (“wind seeds 3–6” and “wind seeds 7–10”). Fatigue is the damage-equivalent load from rainflow counting [13, 14]. The objective is the metric set of [4] as one scalar,

$$J = \frac{1}{4} \sum_{m \in \{P, \omega, T, B\}} \text{clip}_{\pm 100}(\Delta m [\%]) - 20 \max(0, E_{\text{loss}} [\%] - 1), \quad (1)$$

the mean of the per-episode-clipped percentage reductions, relative to paired GSPI, of power MSE and generator-speed MSE on above-rated steps (labelled by hub wind, whatever the controller does) and of tower-base and blade-root fatigue, minus a penalty for energy loss beyond

1 % (Appendix A). Every selection in the loop (fork winner, best checkpoint, rollback) uses J on wind seeds 1–2; every reported number is a held-out evaluation of the best checkpoint. Seeds are paired across arms; differences are reported as mean, wins/ n , the exact two-sided sign-flip permutation p (floor $2/2^n$: 0.0625 at $n = 5$, 0.25 at $n = 3$) and a paired t -test. A seed is *balanced* when all four metrics improve on a held-out set. Arms with $n = 3$ are proofs of concept and are labelled as such.

2.2 The residual and its two bases

Two PPO [15] agents—one per operating region, selected by an oracle rule on ROSCO’s native pitch command—add a bounded collective-pitch offset ($|\Delta\beta| \leq 0.05$ rad, second-order damped) to the base command; the pitch limits apply to the sum. Observations: generator-speed error and its rate, measured pitch, hub wind speed, blade-root out-of-plane moment, tower-top fore-aft acceleration, and on the MPC base the base’s target. $\gamma = 0.998$, GAE $\lambda = 0.98$ [16], 64×64 tanh networks, 300 training episodes per run; the critic’s regression target is standardised by a running estimate of the return scale [17]. The per-step reward is region-conditional,

$$r_t = \mathbf{1}[\text{Region } 2] w_P \left(\frac{P}{P_{\text{GSPI}}} - 1 \right) - \mathbf{1}[v_{\text{hub}} > v_{\text{rated}}] w_\omega \left(\frac{\delta\omega}{0.005} \right)^2 - \lambda_T \ell_T - \lambda_B \ell_B - \lambda_A (\Delta\beta/\kappa)^2, \quad (2)$$

with fatigue proxies ℓ_T, ℓ_B (growth of the 10s peak-to-peak range of tower acceleration and of blade-root moment, ≈ 1 per step under GSPI) and the speed term gated by the wind label, the subset on which the objective’s regulation metrics are scored (Appendix B). Its six coefficients and the two residual bounds are the *knobs*; their default is tuned once on wind seeds 1–2 by J (Appendix A) and is the same on both bases.

GSPI base: the residual sits on ROSCO’s PI command. *MPC base*: a three-state LPV-MPC (rotor, first tower fore-aft mode; linearised every 0.1s from the C_P/C_T tables; horizon 2s; OSQP [18]; no wind preview) runs inside the environment, the residual is added to its target, and its target is in the observation. Its cost terms are scaled to $O(1)$, which is what makes it the strongest controller on J : with the speed error normalised by rated speed instead, the tower term out-weighs regulation by four orders of magnitude and the optimum feathers the rotor, reproducing the published verdict that MPC “wins regulation and loses tower fatigue” (Appendix C). With zero residual the controller is the MPC itself, so the episode-0 policy is the MPC. The MPC is also run with the power and thrust coefficients scaled by 0.95 in its internal model only (the plant is unchanged), the case in which the residual has a model error to repair. All MPC-alone numbers are the MPC as the base controller with a zero residual, evaluated through the same code path as every residual run (Appendix C records why this matters).

2.3 The supervision loop, its levers and their controls

Every 30 episodes the policy is evaluated on wind seeds 1–2 and a supervisor may act. The agent (an OpenAI-compatible reasoning model in JSON mode, about five calls per run) receives the training diagnostics, the evaluation, the decision history with outcomes and the objective’s definition, and returns $K = 3$ candidates. Each candidate is trained for a 30-episode fork from the same checkpoint, the forks are evaluated on wind seeds 1–2, and the best fork continues (*fork verification*). A guardrail rolls the run back to its best checkpoint when the objective drops by more than five points or energy loss exceeds 1 %. Levers: the six **weights** of (2), with log-uniform random search as control; the PPO **hyper-parameters** (learning rates, GAE λ , entropy, clip), with random search as control; the **reward expression**, a sandboxed Python expression over (region, wind label, speed error, power ratio, both fatigue proxies, actuation) that replaces (2) in the manner of Eureka [5], with as control a random *structural* search over expressions drawn from a fixed grammar of the same term shapes with log-uniform coefficients, passed through the same loop; and both namespaces in **one agent** (hyper-parameters and reward

Table 1: The supervision study on the GSPI base (held-out J , mean $\pm$ s.d.; composition on wind seeds 3–6 as power MSE / speed MSE / tower fatigue / blade fatigue, % reduction vs. GSPI; “balanced” counts seeds on which all four metrics improve; paired differences against the tuned fixed reward with wins and exact p on wind seeds 3–6 / 7–10).

arm	n	J seeds 3–6	J seeds 7–10	P / ω / T / B	balanced	ΔJ vs. fixed (wins, exact p)
fixed reward (tuned)	5	4.86 ± 2.76	5.50 ± 2.09	17.4 / 19.0 / -17.6 / 0.7	0/5	—
LLM hyper-parameters	3	5.26 ± 2.39	7.13 ± 3.05	17.1 / 20.7 / -18.1 / 1.2	0/3	-0.5 / $+0.3$ (2/3, 1.0)
random hyper-parameters	3	4.93 ± 2.02	5.96 ± 2.17	15.1 / 17.0 / -13.2 / 0.8	0/3	-0.8 / -0.9 (2/3, 1.0)
LLM hyper-parameters + reward, one agent	5	6.37 ± 2.94	8.65 ± 2.60	15.9 / 16.3 / -7.6 / 0.9	0/5	$+1.5$ / $+3.2$ (4/5, 0.31 / 0.13)
LLM reward expression	5	5.30 ± 4.17	6.04 ± 4.28	6.8 / 10.1 / $+1.5$ / 2.7	3/5	$+0.4$ / $+0.5$ (3/5, 0.75 / 0.69)
LLM reward expression, told a load-priority objective	5	7.00 ± 2.12	8.33 ± 2.38	9.5 / 14.6 / $+0.9$ / 3.0	3/5	$+2.1$ / $+2.8$ (4/5, 0.31 / 0.19)
random reward structure, same loop	3	6.13 ± 0.52	7.06 ± 1.35	9.5 / 10.5 / $+2.1$ / 2.5	1/3	$+0.4$ / $+0.3$ (2/3, 1/3; ≈ 1)
LLM reward expression, single-shot, no loop	5	4.97 ± 1.90	6.66 ± 2.75	11.6 / 13.3 / -5.5 / 0.5	1/5	$+0.1$ / $+1.2$ (2/5, 3/5; ≈ 1)

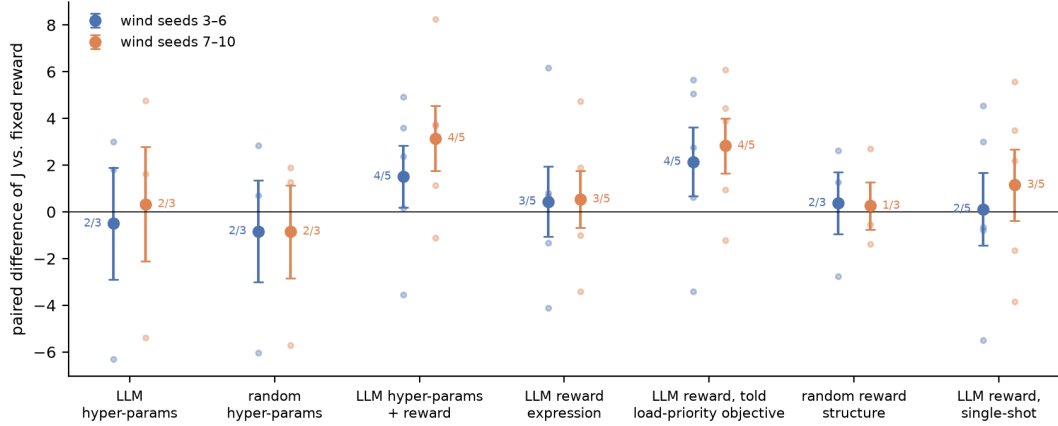


Figure 1: Seed-paired difference of held-out J between each supervised arm and the tuned fixed reward on the GSPI base, both held-out wind sets (mean $\pm$ s.e., dots are seeds, labels are wins/ n). No arm is significantly above zero.

expression). Fixed-knob training under the same guardrail is the *fixed-reward* baseline. Loop ablations: the agent asked once at the first decision and applied without a fork (*single-shot*); the agent told a different objective than the one that selects (a load-priority objective with speed regulation as a penalised constraint; only what the agent is *told* changes, J still selects every fork and checkpoint).

3 Results

3.1 Levers: numeric ones are worth nothing, the reward’s shape changes the composition

On the GSPI base the residual is the whole controller’s improvement, so the levers have room to matter. Table 1 and Figs. 1–2 give the outcome, all on the tuned default and both held-out sets.

Numeric levers. The hyper-parameter agent is worth -0.5 / $+0.3$ and random hyper-parameter search -0.8 / -0.9 : the two proposers are indistinguishable and neither moves the fixed reward. The weight search (Appendix A) raises J by about two points over an untuned start—that is the tuning the default already contains—and never keeps a candidate with a heavier fatigue weight: fatigue responds over tens of episodes, regulation within one fork, so a 30-episode verification horizon is myopic for that knob (Fig. 5b), and a sweep of the tower weight confirms the balanced region is not on that axis at all ($\lambda_T = 3, 10, 30$: 1.4, -2.9 , no learning).

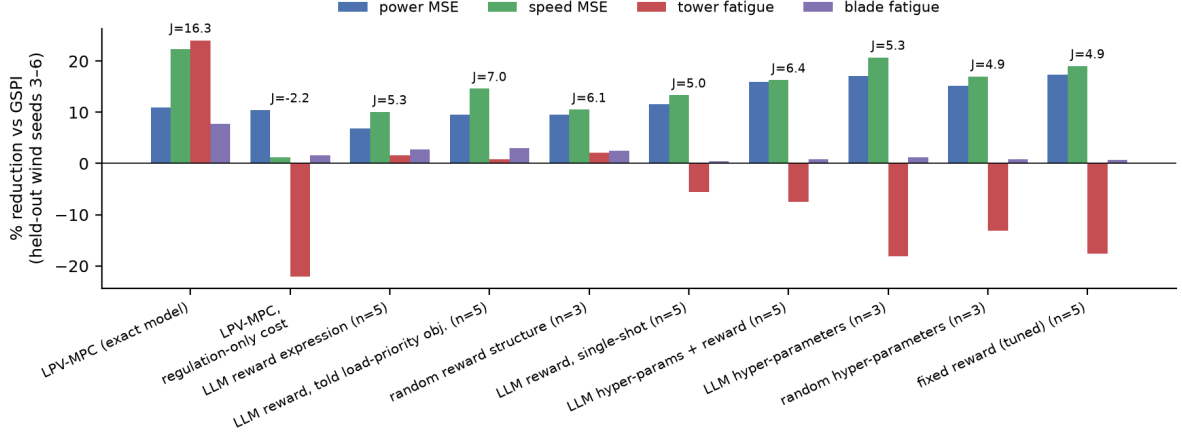


Figure 2: What each controller’s held-out J is made of (wind seeds 3–6, arm means), with the MPC rows for scale. Every fixed-reward and hyper-parameter arm buys regulation with tower fatigue; the reward-expression agent, the random draw from its vocabulary and the correctly scaled MPC are the rows with every term positive.

The reward’s shape. Every fixed-reward and hyper-parameter arm buys regulation with tower fatigue (-13 to -18%), the opposite of what a residual load controller is for. The reward-expression agent is the only lever whose mean tower fatigue improves and the only one with balanced seeds (3/5); its balanced seeds share a *saturated* speed term $\tanh(|\delta\omega|/0.005)$ on the wind-labelled subset and centred, bounded load terms $\tanh(\ell - 1)$ that penalise only above-baseline load (Appendix D)—a change of reward shape that no weight of (2) can express. Its effect on the level of J is $+0.4 / +0.5$ (3/5, $p = 0.75 / 0.69$), inside seed noise. Combining the reward lever with the hyper-parameter lever in one agent raises the level ($+1.5 / +3.2$, 4/5, $p = 0.31 / 0.13$) but never the composition (0/5 balanced): it buys regulation the way the hyper-parameter arms do.

3.2 The loop is necessary, the prompt is not, and the prior is a vocabulary

Loop. Asked once at the first decision and applied without a fork, the same agent is the fixed reward ($+0.1 / +1.2$, 1/5 balanced). Told a load-priority objective, with J still selecting its forks, it writes the same shapes and reaches the balanced solution as often (3/5), with the largest paired gain of any arm ($+2.1 / +2.8$, 4/5, $p = 0.31 / 0.19$); its fork records show it choosing between candidates whose told objective and J disagreed, and J won each time because J selects. Pooled over the two looped reward-expression arms the balanced solution appears in 6 of 10 seeds, in 1 of 5 without the loop and in 0 of 5 fixed-reward seeds.

Prior. A random search over expressions drawn from a fixed grammar of the term families the agent uses, verified by the same loop, lands where the agent lands: $6.13 \pm 0.52 / 7.06 \pm 1.35$, a positive mean tower fatigue, the balanced solution in 1 of 3 seeds, and the saturated forms kept. Once the vocabulary is written down, verification against the objective finds the balanced solution without an LLM; whether the agent finds it more reliably (6/10 against 1/3) is not separable at these seed counts. The agent’s contribution, then, is the vocabulary: it wrote, unprompted, the saturated and centred forms that the grammar of the control was built from.

3.3 What the compositions look like: one held-out episode

Fig. 3 shows the two compositions in time on one held-out episode chosen by a stated rule. On the GSPI base both residuals hold power closer to rated than GSPI (power MSE 1.9 and 2.1

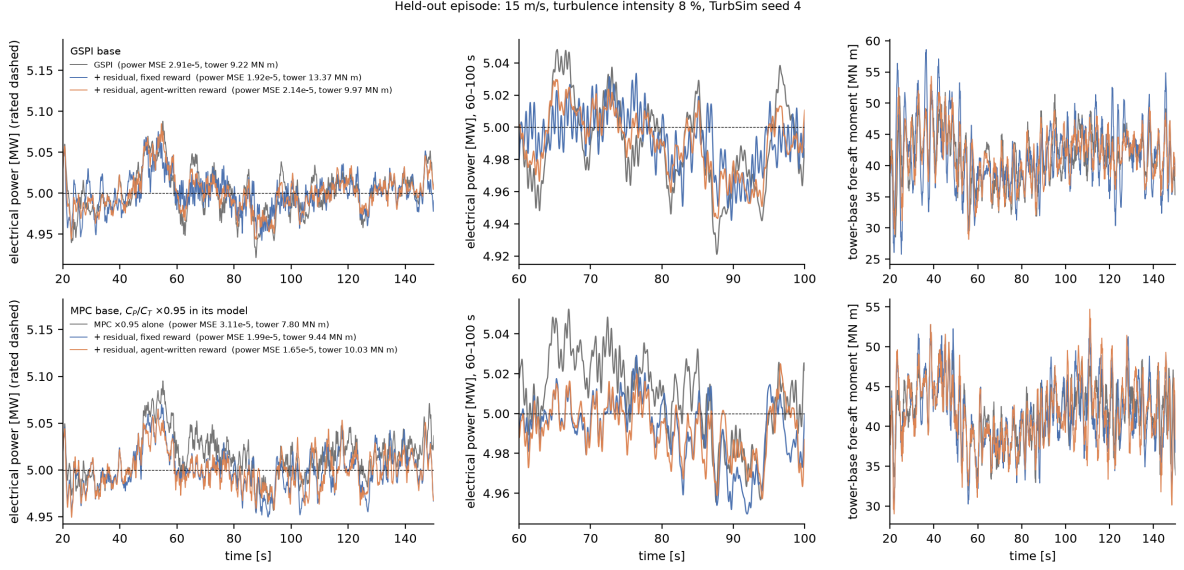


Figure 3: One held-out above-rated episode, both bases: how close each controller holds power to rated, and what it costs the tower. Top: GSPI, the fixed-reward residual and the agent-reward residual on the GSPI base (one seed each, the seed nearest its arm’s mean). Bottom: the MPC with the 5 % model error alone and with the two residuals. Left: electrical power over the scored window (rated dashed); middle: the same, 60–100 s; right: tower-base fore-aft moment. The legend gives the episode’s power MSE on above-rated steps and its tower-base fatigue; above rated the generator torque is constant, so the generator-speed trace is the power trace scaled and its MSE is the same number. The episode is the 15 m/s realisation on which the two GSPI-base arms’ tower-fatigue difference is closest to its mean over the four held-out realisations.

Table 2: The residual on the MPC base (held-out; mean $\pm$ s.d.; composition on wind seeds 3–6; $n = 3$, proof of concept). The MPC-alone rows are the MPC as the base controller with a zero residual. Energy is within $\pm 0.2\%$ of GSPI in every row.

controller	n	J seeds 3–6	J seeds 7–10	$P/\omega/T/B$ (seeds 3–6)	ΔJ vs. MPC alone (wins)
LPV-MPC alone, exact model	—	16.28	17.22	11.0 / 22.3 / 24.1 / 7.8	—
MPC + residual, fixed reward	3	16.60 ± 0.78	18.06 ± 1.10	12.1 / 23.6 / 24.2 / 6.4	+0.3 / +0.8 (2/3, 2/3)
MPC + residual, LLM-written reward	3	17.23 ± 1.30	18.84 ± 1.04	13.4 / 24.7 / 23.6 / 7.2	+1.0 / +1.6 (2/3, 3/3)
LPV-MPC alone, $C_P/C_T \times 0.95$ in its model	—	8.53	3.57	1.5 / 4.7 / 23.4 / 4.5	—
$\times 0.95$ MPC + residual, fixed reward	3	14.17 ± 0.82	15.35 ± 0.68	18.6 / 27.6 / 6.7 / 3.8	+5.6 / +11.8 (3/3, 3/3)
$\times 0.95$ MPC + residual, LLM-written reward	3	12.66 ± 5.13	18.25 ± 2.52	13.0 / 22.7 / 9.3 / 5.7	+4.1 / +14.7 (2/3, 3/3)

against 2.9×10^{-5}); the fixed-reward residual buys it by driving the tower harder (tower-base fatigue 13.4 against 9.2 MN m), the agent-reward residual does not (10.0), which is the difference between -17.6% and $+1.5\%$ tower fatigue in Table 1 at the same J . On the MPC base with the model error, the MPC alone under-pitches for the plant it actually controls and its speed sits above rated for tens of seconds; both residuals correct it, at a small cost in tower fatigue, and the two residuals are indistinguishable.

3.4 The strong base: little left to add, a model to repair, and the same verdict on the agent

The MPC is positive on every metric on its own (Table 2), three times the fixed-reward residual on the GSPI base, and the residual has correspondingly little left to do: +0.3 with the fixed reward and +1.0 with the agent’s, both 2/3 seeds, both within the seed spread; in half of the exact-model runs the best checkpoint is the initial one, i.e. the MPC itself (Appendix E, Fig. 7). When the MPC’s power and thrust coefficients are 5 % low in its model—alone at 8.5 / 3.6, its regulation degraded and its power MSE worse than GSPI on one held-out set—the residual

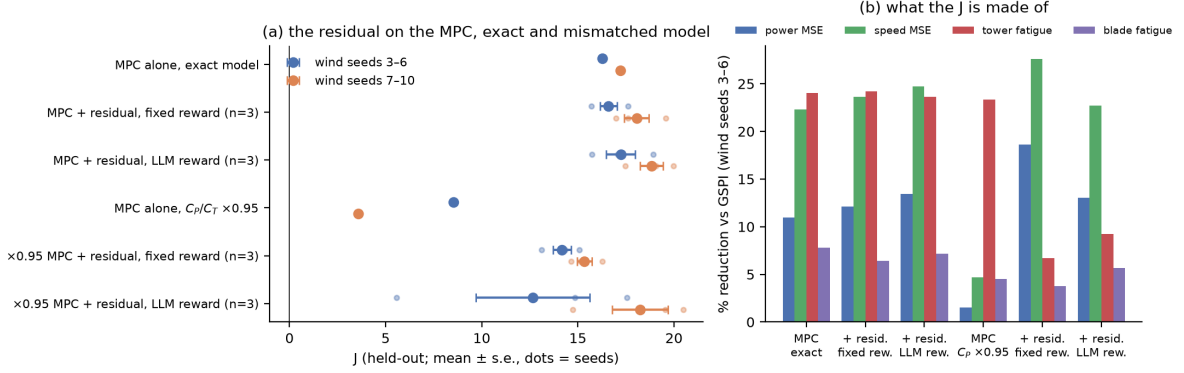


Figure 4: Residual on the MPC base. (a) Held-out J on both wind sets for the MPC alone and with the two residuals, with an exact model and with the power/thrust coefficients 5% low in the MPC’s model (mean $\pm$ s.e., dots are seeds). (b) The four terms of J on wind seeds 3–6 for the same rows.

repairs it: the fixed-reward arm reaches 14.2 / 15.4 in every seed and the agent arm 12.7 / 18.3 with one poor seed, above the exact-model MPC alone on wind seeds 7–10. The residual learns the part of the plant the model gets wrong, on wind realisations it has never seen. Between the fixed reward and the agent’s, the difference is +0.6 / +0.8 (2/3) on the exact model and $-1.5 / +2.9$ (one poor seed) on the mismatched one: on the strong base, as on the weak one, the agent does not move the level, and here the composition is the MPC’s already.

4 What the agent contributes, and rules for finding out

The evidence on both bases points one way. The agent’s demonstrable contribution is *structural*: a vocabulary of reward shapes (saturated regulation terms, centred bounded load terms) that reaches a solution class no weight search reaches. It is *realised by verification*: the same vocabulary drawn at random and verified by the same loop reaches the same class; the agent without the loop reaches nothing; the agent told the wrong objective and selected by the right one reaches the class. It is *not a level effect*: no paired difference against the fixed reward is significant on either base, and on the strongest base the residual itself has little to add. We distil this, and the defects that mattered more than the agent, into rules.

1. **Verification against the true objective is the guarantee; the prompt is not.** Select on the objective you report; the agent’s stated objective is a prior, not a constraint.
2. **Put the agent on structural decisions; give numeric ones to random search, or to nothing.** No numeric lever, LLM or random, moved a tuned default; the reward’s shape did.
3. **Publish the grammar.** Once the agent’s vocabulary of shapes is written down, a verified search over it is a reproducible, LLM-free control—and the honest baseline for the next agent.
4. **One lever per agent.** The combined agent gained a little level and lost the composition effect.
5. **Fix the protocol before crediting the agent.** Four defects each moved the comparison by more than any agent: a reward speed term gated on a different subset than the objective’s metric (Appendix B), an unnormalised critic target, an MPC cost with terms four orders of magnitude apart, and a reference controller evaluated through the learner’s action channel instead of as itself (Appendix C).

6. **Report composition and reliability, not only means.** The agent’s contribution—a solution class, in a fraction of seeds—is invisible in a table of mean J .
7. **Expect the agent’s headroom to shrink with the base.** On the strongest base the residual, and therefore its supervisor, had little left to do; the agent’s place is where the base is weak or its model wrong.

5 Limitations

Seed budgets are five for the core GSPI-base arms and three for the controls and the MPC-base arms; no agentic difference in J is significant, and the composition effect (3/5 balanced) rests on few seeds. One plant, one training turbulence intensity, two held-out sets of four realisations, one reasoning model with one prompt family. The MPC has no preview, a crude tower-state estimator and a two-second horizon; the model error studied is a scalar scaling of the aerodynamic coefficients, not an identified model. The residual policy observes the simulator’s hub wind where the MPC uses ROSCO’s estimate. An individual-pitch extension of the same supervisor was attempted and did not win on any of its levers; the boundary of the present claims is collective pitch.

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A Objective, metrics and the tuned default

The four metrics of (1) are computed per episode against the paired GSPI run on the same wind realisation: power MSE and generator-speed MSE relative to rated on the steps where hub wind exceeds rated (the 8 m/s episodes contribute no regulation term), and damage-equivalent loads of tower-base fore-aft moment and blade-root out-of-plane moment over all episodes. Each episode’s reduction is clipped to $\pm 100\%$ before averaging so that a single blown regulation episode cannot own the objective. Energy is the total over all episodes.

The tuned default of (2) is the per-knob geometric median of the best checkpoints of a three-seed random-search arm over the six knobs on the GSPI base (fork-verified, J -selected on wind seeds 1–2): $w_P = 220$, $w_\omega = 25.2$, $\lambda_T = 1$, $\lambda_B = 1.31$, residual bounds 0.074 rad (below rated) and 0.05 rad (above). That search raised held-out J from $4.60 \pm 2.38 / 5.96 \pm 2.02$ (untuned) to $6.70 \pm 2.12 / 7.89 \pm 2.21$ but never kept a candidate with $\lambda_T > 1$ (Fig. 5b). The tower-weight sweep from the tuned default (seed 0, wind seeds 3–6 / 7–10): $\lambda_T = 1$: 5.0 / 5.9; 3: 1.4 / 3.7; 10: $-2.9 / -3.3$; 30: the policy never left its initial state. The same default is used unchanged on the MPC base.

B Why the reward’s speed term is gated by the wind label

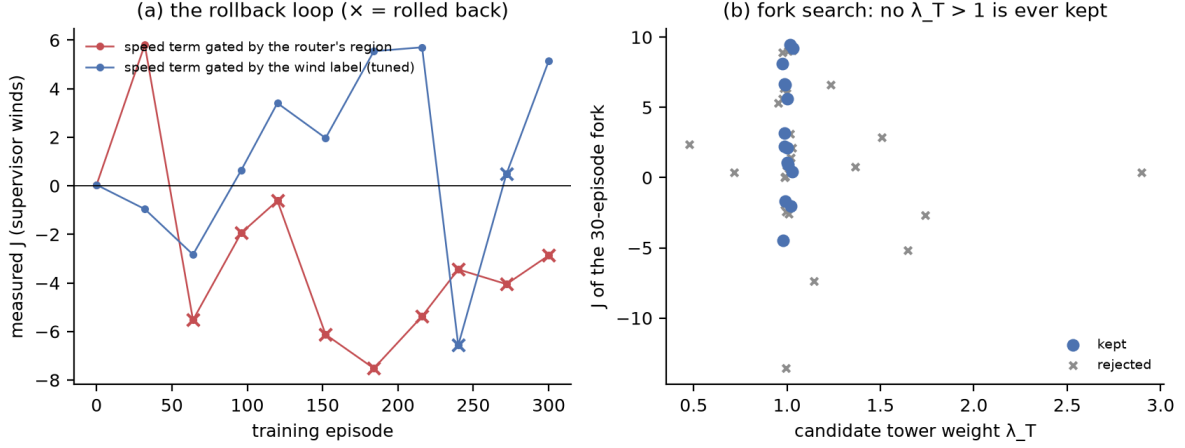


Figure 5: (a) Measured J on the supervisor winds at every decision of a fixed-reward run whose speed term is gated by the router’s region (red) and of the tuned default gated by the wind label (blue); crosses mark rollbacks. (b) Every fork candidate’s tower weight against the J of its 30-episode fork in the weight search; no candidate with $\lambda_T > 1$ is ever kept.

The objective’s regulation metrics are scored on the steps where hub wind exceeds rated, whatever the controller does. If the reward’s speed term is instead gated by the router’s region, the transition steps at 12.5 m/s—router “below rated”, objective “above”—carry no speed penalty, and the policy is rewarded there for load reduction that the objective scores as regulation loss. Under that gating every fixed-reward run found an early optimum and then rolled back at every later decision (Fig. 5a): held-out J $1.59 \pm 0.87 / 0.97 \pm 1.37$ ($n = 3$). On that default, hyper-parameter supervision appears valuable (+5.7 / +8.0 for the LLM, +5.8 / +7.5 for random search, both by lengthening the credit window, GAE λ 0.98→0.995, and slowing the actor). Gating the speed term by the wind label removes the loop without any agent, and on that default the same hyper-parameter arms are worth nothing (§3.1). The aligned gating is used throughout the main text; the lesson is rule 5.

C The MPC: cost scaling, selection, evaluation channel, model error

The controller model is $J\dot{\omega} = T_{\text{aero}}(v - \dot{x}, \omega, \beta) - T_{\text{gen}}(\omega)$, $m\ddot{x} = F_{\text{thrust}}(v - \dot{x}, \omega, \beta) - c\dot{x} - kx$ with the first tower fore-aft mode at 0.324 Hz, modal mass 437 t and structural damping ratio 0.01, linearised at the operating point from the C_P/C_T tables every 0.1 s and discretised exactly; tower states are estimated by leaky double integration of the measured nacelle acceleration; the wind input is ROSCO’s estimate low-passed at 0.35 rad/s; torque is native ROSCO. The condensed QP over $N = 20$ steps minimises $\sum_k q (\delta\omega_k / (0.005\omega_{\text{rated}}))^2 + q_t (\dot{x}_k / 0.2)^2 + r (\Delta\beta_k / 0.002)^2$ subject to the live pitch floor, the pitch limit and the rate limit, 3.6 ms per solve. Selection on wind seeds 1–2 by J over $N \in \{20, 40\} \times r \in \{0.1, 0.3, 1, 3\} \times q_t \in \{0, 0.1, 0.3, 1, 3\} \times$ two wind-filter corners (82 points) gives $N = 20$, $r = 0.3$, $q_t = 3$; $N = 40$ is worse everywhere.

Cost scaling. With the speed error normalised by rated speed and the pitch increment by 0.1 rad (the natural first choice), the speed term is $O(10^{-5})$ at a typical error of 0.5 % while the tower term is $O(1)$; any tower weight above zero then out-weighs regulation by four orders of magnitude and the optimum feathers the rotor, so the best such MPC has no tower term and

loses tower fatigue—the published verdict (Table 3, regulation-only row). Scaling each term to $O(1)$ at its typical value makes the same three-state MPC positive on every metric.

Evaluation channel. An earlier version of this study evaluated the MPC by sending its command through the residual channel of the RL agents (as an offset on ROSCO’s command with the channel’s bounds off). That channel has a rule written for the learned residual—below rated, by the router’s region, the offset is non-negative—which zeroed the MPC’s negative offsets during the router’s transition hold at 12.5 m/s and produced pitch discontinuities: on one held-out episode the tower-base fatigue was 11.6 MN m through the channel and 7.3 MN m for the same MPC run as the base controller, against 19.4 for GSPI. Every MPC-alone number in this paper is therefore the MPC run as the base controller with a zero residual, i.e. through exactly the code path of the residual runs; the lesson is rule 5.

Model error. As the base controller, with full authority, the MPC is sharper in its model sensitivity than a bounded controller would be (Table 3, Fig. 6): tower frequency $\pm 10\%$ and modal mass -20% cost 3–4 points, modal mass $+20\%$ costs nothing, off-design turbulence (14%, 22%) costs 2–4 and 18 m/s is easier; but the power and thrust coefficients 5% low or high take it to 8.5 and 5.7, and $\pm 15\%$ to -24 , the regulation terms collapsing. An identified aerodynamic model will not be exact; this is what the residual of §3.4 repairs.

Table 3: LPV-MPC as the base controller (J -selected weights) on the held-out sets, and under perturbation of its internal model only and off-design winds (wind seeds 3–6 unless noted).

variant	J seeds 3–6	J seeds 7–10	P / ω / T / B (seeds 3–6)	note
exact model	16.28	17.22	11.0 / 22.3 / 24.1 / 7.8	base of §3.4
$C_P/C_T \times 0.95$	8.53	3.57	1.5 / 4.7 / 23.4 / 4.5	base of §3.4
regulation-only cost ($q_t = 0$, unscaled references)	-2.19	-6.20	10.5 / 1.2 / -22.0 / 1.6	the published verdict
$C_P/C_T \times 0.85 / \times 1.05 / \times 1.15$	-24.0 / 5.7 / -24.6			regulation collapses at $\pm 15\%$
tower frequency $\times 0.9 / \times 1.1$	12.0 / 13.4			
modal mass $\times 0.8 / \times 1.2$	13.3 / 17.6			
turbulence intensity 14% (seeds 3–4) / 22% (15 m/s, seeds 3–4) / 18 m/s (seeds 3–4)	12.6 / 14.2 / 23.1			off-design winds

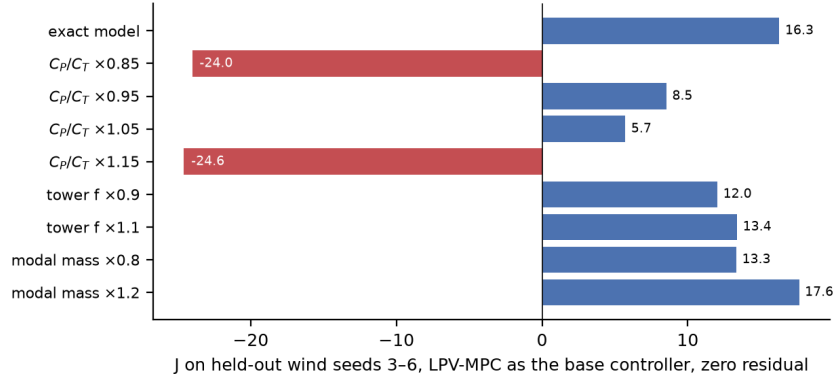


Figure 6: The MPC is conditional on its aerodynamic model. Held-out J of the MPC as the base controller when only its internal model is perturbed; the plant is unchanged.

D Reward expressions written by the agent

Best-checkpoint expressions of the reward-expression agent on the GSPI base (variables: **region** = router region, **region_w** = wind label, **d_wg** = relative speed error, **p_ratio** = power over paired GSPI power, **load_t**, **load_b** = fatigue proxies, **act** = actuation cost). Seed 0, told J :

```

180*tanh(15*(p_ratio-1))                                if region==0
-40*sqrt(tanh(abs(d_wg)/.005)**2 + tanh(60*abs(p_ratio-1))**2)  if region_w==1
-.9*tanh(log((2+load_t+load_b)/4)) - .1*tanh(act)

```

Seed 1, told the load-priority objective:

```

155*tanh(p_ratio-1)                                if region==0
-26*tanh(abs(d_wg)/.005)                            if region_w==1
-(1.4 if region==0 else .5)*tanh(load_t)
-(1.0 if region==0 else 1.5)*tanh(load_b) - .1*tanh(act)

```

Seed 0, told the load-priority objective:

```

130*tanh(6*(p_ratio-1))*(region==0) - 24*tanh(d_wg/.005)**2*(region_w==1)
+ 1.0*tanh((1-load_t)/(1+load_t)) + 1.1*tanh((1-load_b)/(1+load_b)) - .04*tanh(act)

```

The random structural control samples expressions of the same families with log-uniform coefficients spanning the ranges the agent used. Speed terms, in $x = \delta\omega/0.005$: x^2 , $|x|$, $\tanh|x|$, $\tanh(x^2)$, $\sqrt{|x|}$. Load terms, in $\ell \in \{\ell_T, \ell_B\}$: ℓ , $\min(\ell, 3)$, $\tanh(\ell - 1)$, $\tanh \ell$, $\max(0, \ell - 1)$, $\log(1 + \ell)$. An optional power-deviation term on the wind-labelled subset.

E Training-time behaviour and the full controller list

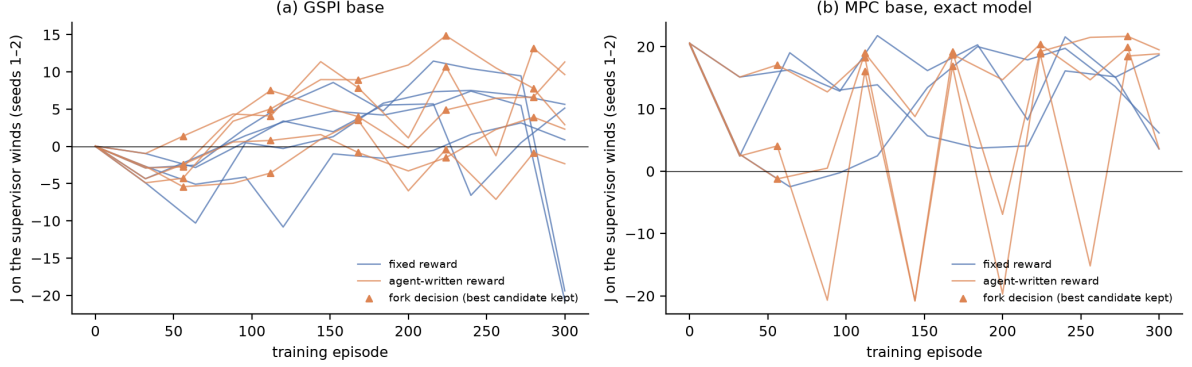


Figure 7: J on the supervisor winds at every evaluation during training, fixed-reward and agent-reward runs, (a) on the GSPI base and (b) on the MPC base with the exact model; triangles mark fork decisions of the agent runs, and the deep dips of the agent runs on the MPC base are evaluations that triggered the guardrail’s rollback. On the MPC base the episode-0 policy is the MPC itself and half of the runs never exceed it.

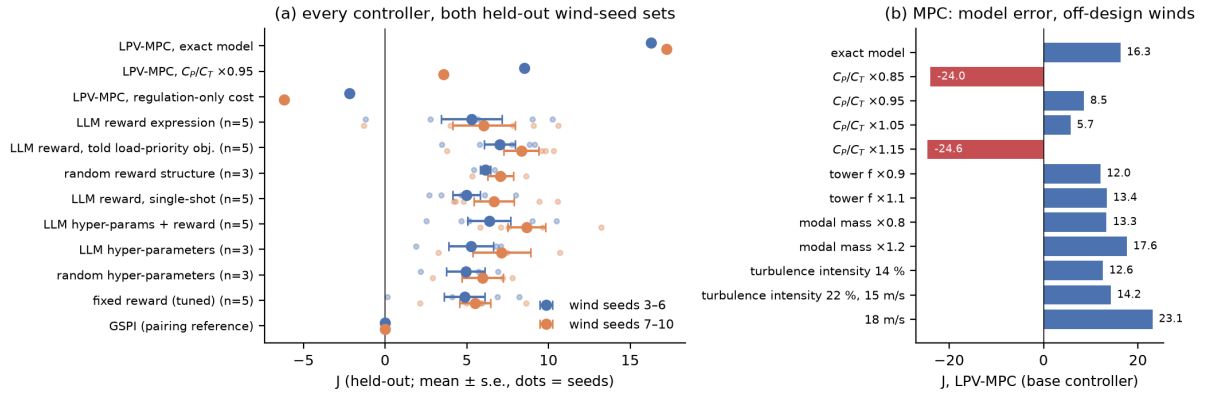


Figure 8: (a) J of every GSPI-base controller and of the MPC (as base controller) on both held-out sets. (b) The MPC under model error and off-design winds (the data of Fig. 6, with the wind classes).

F Compute and reproducibility

One 300-episode training run takes 65–85 minutes on 16 CPU cores (8 OpenFAST workers, two runs in parallel); the MPC in the worker adds 3.6 ms per 0.1 s solve. A supervised run makes about five LLM calls. All runs write rolling checkpoints and can be paused and resumed; every number in this paper is regenerated by the scripts in the repository from the run artefacts, and the figures by a single script.